বাংলাদেশ আর্মি ইন্টারন্যাশনাল ইউনিভার্সিটি অব সায়েন্স এন্ড টেকনোলজি (বিএআইইউএসটি) কুমিল্লা
BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST) CUMILLA

Department of CSE
Level-2, Term-II
Mid Term Examination, Fall 2024
Course Code: GED221
Course Title: Engineering Economics

Time: 1.5Hr
Full Marks: 90

Notes:

a. Answer any *03* (*Three*) of the following questions.
b. Each question carries 30 marks.
c. Figure on the right of each question indicate marks for respective question.

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1. a) Define Economics from welfare perspective. Define scarcity. **[15]**

   b) As an Engineering student, why should you study Economics? **[15]**

2. a)Explain PPF (with figure). Indicate implications of the different points of PPF curve. **[20]**

   b)Analyze the opportunity cost with examples. **[10]**

3. a)Explain factors of production. Which of the factors is the most important? Explain. **[15]**

   b) "Supply is one of the vital forces in an economy"-explain. **[15]**

4. From the following data develop an equation. **[30]**

Period	Price	Quantity
2021	20 tk.	500 units
2022	30 tk.	700 units

From the developed equation prepare a schedule as well as a curve showing price-quantity relationship.

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BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST) CUMILLA

Mid Term Examination, Fall 2024
Department of Computer Science and Engineering (CSE)
Level-2, Term-II
Course Code: MATH 221
Course Title: MATH-IV (Probability and Statistics)

Notes: **Time: 1 hr 30 min.**
Full Marks: 90

a. Figure on the right of each question indicates marks for respective question.
b. **Answer any 03 (Three) of the following 04 (Four) questions.**

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1. a. A hamburger chain found that 75% of all customers` use mustered, 80% use ketchup and 65% use both. Explain the probabilities that ketchup user uses mustard and mustard user uses ketchup? Also interpret a table for join probability. 11

   b. The probability that a certain person will travel from Dhaka to Rajshahi by train is $\frac{5}{9}$ and that subsequently the person will travel from Rjshahi to Dinajpur by bus is $\frac{2}{7}$. Using a probability tree find the probability that the person will travel from Dhaka to Rajshahi by train and then travel to Dinajpur not by bus. 12

   c. The following table gives the no. of days in a 50 day period during which automobile accidents occurred in a city. Estimate the poisson distribution to the data: 07

No. of accidents	0	1	2	3	4
No. of days	21	18	7	3	1

2. a. The probability that a person recovers from a rare blood disease is 0.4. If 5 people are known to have contracted this disease, Compute the probability that (i) Exactly 3 survive (ii) At least 2 survives (iii) At most two survive and (iv) None survive. 15

   b. The marks obtained in English by 40 students in an annual examination are given below. Construct a cumulative frequency and relative cumulative frequency table of the marks obtained. 15
   70, 40, 35, 60, 55, 58, 45, 60, 65, 80, 70, 46, 50, 60, 65, 70, 58, 60, 48, 70, 36, 85, 60, 50, 46, 65, 55, 61, 72, 85, 90, 68, 65, 50, 40, 56, 60, 65, 46, 76.

3. a. Draw frequency polygon and frequency Histogram from the frequency distribution table of heights of 100 students of a college. 11

Heights (in cm.)	141-150	151-160	161-170	171-180	181-190
Frequency	5	16	56	11	12

   b. The frequency distribution table of the marks obtained by the students of a class is as follows. Find the arithmetic mean of the marks. 12

Class interval	25-34	35-44	45-54	55-64	65-74	75-84	85-94
Frequency	5	10	15	20	30	16	4

c. In a winter season, the temperature (in celsius) of the month of January in the district of Srimangal is placed below. Determine the frequency distribution table of the temperature. 07

$14^0, 14^0, 14^0, 13^0, 12^0, 13^0, 10^0, 10^0, 11^0, 12^0, 11^0, 10^0, 9^0, 8^0, 9^0, 11^0, 10^0, 10^0, 8^0,$
$9^0, 7^0, 6^0, 6^0, 6^0, 6^0, 7^0, 8^0, 9^0, 9^0, 8^0, 7^0.$

4. a. The rate of passing in degree Honours class and the number of students of some department of a University are presented in the table below. Compute the mean rate of passing in degree Honours class of those departments of the University. 10

Name of the Dept.	Math	Statistics	English	Bangla	Zoology	Political Science
Rate of passing (%)	70	80	50	90	60	85
No. of students	80	120	100	225	135	300

b. The frequency distribution table of weights (in kg) of 60 students of a class are:

Class interval	45-49	50-54	55-59	60-64	65-69	70-74
Frequency	4	8	10	20	12	6

10

Demonstrate the mode of the data.

c. Determine median from the following frequency distribution table: 10

Time(Sec.)	30-35	36-41	42-47	48-53	54-59	60-65
Frequency	3	10	18	25	8	6

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BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA
Department of Computer Science & Engineering

Level- II Term-II

Midterm Examination, FALL-2024
Course Code: CSE 223
Course Title: Digital Logic Design

Notes:

a. Answer **any 03 (three)** of the following 4 questions. **Time: 1 hour 30 minutes**
Marks: 90

b. Each question carries 30 marks.
c. Figure on the right of each question indicate marks for respective question.
d. Required Postulates and Theorems of Boolean algebra are given in Table 1.

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1. a. What is Complements? Find out the **2's** complement of **$(101100)_2$**. 10
   b. What is parity bit? Explain all the types of it with their corresponding examples. 10
   c. Write a short note on the following terms: 10
      (i) Integrated Circuits.
      (ii) Gray Code.

2. a. What is Register transfer? Show me in a figure how binary information is processed with registers. 10

   OR

   Your ID (Last 4 digit) $+10011011_2+AB_{16} = (?)_8$

   b. Subtract $(1000100)_2$ from $(1010100)_2$ using 1's complement. 10
   c. Convert $32_{10} = (?)_2$ with the help of division-remainder technique. 10

3. a. Using existing laws, proof Distributive Law **(7-b)**. 10
   b. Express the Boolean function, $F = XY + X'Z$ in a product of max-term form. 10
   c. Using K-map, convert the following expression into 10

      $$F(A, B, C, D) = \sum (0, 1, 3, 5, 9, 11, 14)$$

      (i) Sum of Products (SOP) and
      (ii) Product of Sums (POS):

4. a. Reduce the following Boolean expression into minimum number of literals: 10

$$AB + A' CD + A' BD + A' CD' + ABCD$$

b. Simplify the following Boolean function using K-map: 20

$$F(A, B, C, D) = \sum (2, 5, 9, 14)$$

And don't care conditions:

$$d(A, B, C, D) = \sum (0, 7, 12, 13)$$

Table 1:

1	Postulate 2	(a) X+0 = X	(b) X.1 = X
2	Postulate 5	(a) X+X′ = 1	(b) X. X′ = 0
3	Theorem 1	(a) X+X = X	(b) X.X = X
4	Theorem 2	(a) X+1 = 1	(b) X.0 = 0
5	Postulate 3, Commutative	(a) X+Y = Y+X	(b) XY = YX
6	Theorem 4, Associative	(a) X+(Y+Z) = (X+Y)+Z	(b) X(YZ) = (XY)Z
7	Postulate 4, Distributive	(a) X(Y+Z) = XY+XZ	(b) X+YZ = (X+Y) (X+Z)
8	Theorem 5, De-Morgan	(a) (X + Y)′ = X′ Y′	(b) (XY)′ = X′+Y′
9	Theorem 6, Absorption	(a) X+XY = X	(b) X(X+Y) = X

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BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA

Mid Term Examination, Fall 2024

Department of Computer Science and Engineering

Level-2 Term-II

Course Code: CSE 221

Course Title: Computer Algorithms and Complexity

Credit Hour: 3.0

Exam Duration: 1 hour 30 Minutes

Full Marks: 90

Notes:

a. Figure on the right of each question indicates marks for respective question.

b. Answer ***any Three*** questions out of **Four**.

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1. a. Analyze the given code segment and determine the count of each type of statement. 10

```
for (int i = 0; i < 4; i++) {
    for (int j = 0; j < 4; j++) {
        cout << "i=" << i << ", j=" << j << endl;
        cout << "i=" << i << ", j=" << j << endl;
    }
}
```

b. Construct a Step Table for the following algorithm. 10

```
Algorithm sum(a,n)
{
        s:=0;
        for i:=1 to n do
                s:=s+a[i]
        return s;
}
```

c. Explain the sequence of moves required to solve the Tower of Hanoi problem with 3 disks. Use a source peg (A), an auxiliary peg (B), and a destination peg (C). Illustrate each step involved in the process. 10

2. a. Write the pseudo-code for inserting a value into a Binary Search Tree (BST). Ensure the pseudo-code maintains the BST property, where the left subtree contains values smaller than the node and the right subtree contains values greater than the node. 15

b. Build a max-heap tree for the set of elements {55, 30, 75, 20, 65, 90, 45} step by step, explaining each insertion and adjustment made to maintain the heap property. 15

3. a. Given a sorted array arr = {5, 10, 15, 20, 25, 30, 35} and a target value x = 20, write the pseudocode for the recursive binary search algorithm, showing the values of low, high, and mid in each recursive call. 10

   b. Given the array arr = {8, 3, 1, 7, 0, 10, 2}, demonstrate step-by-step how the QuickSort algorithm sorts this array using the first element as the pivot. 10

   c. Describe the step-by-step process of the Bubble Sort algorithm for sorting an array 8, 4, 6, 9, 2, 3, 1 in ascending order. 10

4. a. Write the pseudocode for the merge sort algorithm and explain how the divide-and-conquer strategy is applied. 12

   b. Briefly describe how the Insertion Sort algorithm works, emphasizing it's step-by-step process of building a sorted array. 12

   c. Define the following types of graphs with proper diagram: 6

      i) Regular Graph
      ii) Cyclic Graph
      iii) Complete Graph.

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BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA

Mid Term Examination, Fall 2024

Department of Computer Science and Engineering

Level-2 Term-II

Course Code: CSE-225

Course Title: Numerical methods

Credit Hour: 03

Exam Duration: 1 Hour 30 Minutes

Full Marks: 90

Notes:

a. Figure on the right of each question indicate marks for respective question.

b. Answer ***any THREE*** questions out of ***FOUR***.

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1. a. What is accuracy? How is it affected during the process of numerical computing? [10]

   b. Add the numbers 0.735916 E4 and 0.635742 E6 using Computer Arithmetic algorithm. [10]

   c. Find the roundoff error in storing the number 752.6835 using a four-digit mantissa. [10]

2. a. Explain the difference between the Regula Falsi method, the Bisection method and the Newton Raphson method in root-finding techniques. [10]

   b. Use the Bisection Method to approximate the root of $f(x) = x^3 - 4x - 9$ within the interval [2,3] up to two decimal points. [20]

3. a. For root calculations, why does the False Position method converge more quickly than the Bisection method? Give an explanation in your own words. [10]

   b. Find a root for the equation $2e^x \sin x = 3$ using the False Position method and correct it to three decimal places with three iterations. [20]

4. a. Is the Newton Raphson method always convergent? Write down the steps for finding the root of an equation using Newton Raphson method. [10]

   b. Find a real root of the equation $-4x + \cos x + 2 = 0$, by Newton Raphson method up to four decimal places, assuming $x_0 = 0.5$. [20]

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BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST) CUMILLA

Mid Term Examination, Fall 2024
(RETAKE)
Department of Computer Science and Engineering (CSE)
Level- 2, Term-II
Course Code: MATH 247
Course Title: Mathematics -IV
Credit Hour: 03

Notes:

a. Figure on the right of each question indicate marks for respective question. **Time : 1 hr 30 mins**
b. **Answer any three (03) of the following four (04) questions.** **Full Marks: 90**

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1. a. Find the solution of the Linear differential equation $\sec^2 x \tan y\, dx + \sec^2 y \tan x\, dy = 0$. 10

   b. Explain the differential equation $(x^2 + y^2)\frac{dy}{dx} = xy$ by the method of Homogeneous. 12

   c. Express the Ordinary Differential equation by eliminating the constant from $y = a\cos(mx + b)$, where a and b are arbitrary constant. 08

2. a. If $(x+iy)^{\frac{1}{3}} = a + ib$, where $x, y, a, b \in R$ then show that $\frac{x}{a} - \frac{y}{b} = -2(a^2 + b^2)$. 12

   b. Identify the region in z-plane represented by the set of points: $|z-1| \geq 3$. 10

   c. Solve the quadratic equation $2z^2 + 3iz + 2 = 0$. 08

3. a. Find the derivative of $f(z) = 3z^{-2}$ at the point $z = 2$. 15

   b. Show that the function $f(z) = u + iv = \frac{xy^2(x+iy)}{(x^2+y^4)}, z \neq 0$, where $f(0) = 0, z = 0$ satisfied the Cauchy-Riemann equations at the origin. 15

4. a. Show that $\varphi(x,y) = \frac{1}{2}\log(x^2 + y^2)$ is a harmonic function. Also find the harmonic conjugate of this function such that $f(z) = \varphi + i\psi$ is analytic. 15

   b. Determine the singular point of the function $f(z) = \frac{1}{(z+1)(z-i)(z^2-1)}$ and also find where the given function $f(z)$ is analytic. 15